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Clinical study

Identification of clinical characteristics and factors predicting favorable treatment outcomes in elderly patients with traumatic brain injury

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ABSTRACT

Age is an important prognostic factor for patients with traumatic brain injury (TBI), and worse treatment outcomes have been reported in elderly patients. Therefore, proper treatment is needed for the increasing number of elderly patients with TBI. We aimed to analyze predictive factors of favorable treatment outcomes (FTO) in elderly patients. Clinical and radiological data from 493 patients with TBI who visited a single institute were retrospectively collected from January 2014 to December 2015. We compared the characteristics of the elderly group (individuals above 65 years) and younger group (16–65 years). We analyzed the characteristics and outcomes in both groups and the factors related to the Glasgow outcome scale-extended score at 6 months after injury in 170 elderly patients. The treatment outcomes were dichotomized into favorable and unfavorable groups. In the elderly group, the proportion of female patients and the incidence of subdural hemorrhage (SDH) were higher than in the younger group. Among the 170 elderly patients, 62 had pure SDH, and 21 of the 62 with pure SDH had undergone surgical treatment. Compared with other types of intracranial hemorrhage, FTO was as high as 85.5%, and mortality was as low as 11.3% in patients with pure SDH. High initial Glasgow coma scale score, low injury severity score, and normal pupillary reflex were significantly related to FTO in multivariate analysis. Therefore, active therapeutic strategies, including surgery should be considered for elderly patients with pure SDH without intra-parenchymal injury.

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1. Introduction

The speed of population aging has significantly increased in the last several decades [1]. The proportion of individuals aged 65 years and older has doubled from 7% in 1950 to 14% in 2010 [1]. In the United States, the global life expectancy is expected to increase from 65 years of age in 1990 to 83 years of age in 2100, and the elderly population is increasing worldwide [1,2].

In general, subdural hemorrhage (SDH) is more common in elderly patients after head trauma, but some patients show no obvious trauma. Subdural hemorrhage is a hematoma between the dura and arachnoid membranes that occurs due to the rupture of the bridging veins; as age increases, brain volume decreases, and the subdural space expands. Acute SDH is increasing intensively with the increase in the elderly population, resulting mainly from falls and motor vehicle accidents that cause traumatic brain injury (TBI) and acute SDH [3]. Elderly patients with SDH may present

with various symptoms ranging from a mild headache to unconsciousness.

As the range of social activities becomes wider, external activities become more frequent, which in turn increases the incidence of traumatic brain injury (TBI). TBI is a common cause of mortality worldwide, [4] especially in elderly people, with a high mortality and morbidity rate also reported for acute SDH in the elderly population [3]. The annual rates of emergency department (ED) visits of TBI by age groups exhibit two peaks: 15–24 years of age and 70 years of age [5,6]. In addition, TBI-related hospitalization and mortality rates are highest among adults 75 years of age or older (339 per 100,000 and 57 per 100,000, respectively) [5]. Age is an important prognostic factor for patients with TBI [7,8]. Shimoda et al. [9] found that age was a major determinant of outcome in elderly patients with TBI and that surgical treatment was ineffective in elderly patients with a Glasgow coma scale (GCS) score of 3–5 points. Considering the socioeconomic costs and complications, the treatment or management of TBI in the elderly (65 years of age or older, according to the World Health Organization) is particularly difficult [10]. Thus, there is an urgent need to improve

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treatment outcomes in this age group and develop optimal treatment strategies for elderly patients with TBI.

In this study, we aimed to study and compare the clinical characteristics of elderly and younger patients with TBI and to investigate the clinical and radiological factors associated with treatment outcome in elderly patients with TBI.

2. Material and methods

2.1. Patient population

We performed a retrospective cohort study in a single institute from January 2014 to December 2015. This study was performed with the approval of the Institutional Review Board of Ajou University Medical Center (AJIRB-MED-MDB-18-002). The committee waived the requirement for written informed consent because of the anonymization of personal medical information.

We included patients who were admitted to the hospital through ED visits within 24 h after head trauma and those who were diagnosed with TBI. Exclusion criteria were: 1) age younger than 15 years, 2) presence of subacute to chronic hemorrhage, 3) presence of diffuse axonal injury, 4) presence of skull fracture alone (excluding intracranial hemorrhage), and 5) loss to 6-month follow-up visits.

2.2. Comparison between elderly and younger patients with TBI and treatment outcome of elderly patients

Clinical data including sex, GCS score (a tool to assess patient consciousness in the ED), injury severity score (ISS, a medical score to assess trauma severity), and abbreviated injury scale score (an anatomically-based injury-coding system) were reviewed [11]. The presence of intracranial hemorrhage, cerebral contusion or brain parenchymal swelling, and skull fractures were reviewed and identified by brain computed tomography scans. Patients were grouped by age: elderly (≥ 65 years) and younger (16–64 years); their clinical characteristics and treatment outcomes (GOS-E at 6 months after injury and 30-day mortality) were analyzed. We identified factors associated with treatment outcome in elderly patients through 30-day mortality and Glasgow outcome scale-extended (GOS-E) [12] at 6 months after injury. The 30-day mortality rates were estimates of deaths within 30 days after injury, and the GOS-E is a global scale of functional outcome that rates patient status into one of eight categories. The outcome of the elderly patients was dichotomized into two groups: favorable (good recovery to moderate disability; GOS-E > 4) and unfavorable (severe disability to death; GOS-E ≤ 4).

2.3. Statistical analysis

SPSS (IBM® SPSS® Statistics, Version 22, Armonk, NY, USA) was used for all statistical analyses. Categorical variables expressed as number (%) were compared by the chi-square test. If the expected count in any cell was not > 5 , Fisher's exact test was used. Statistical significance was set at $p < 0.05$ for all examinations. Logistic regression analysis was used to identify independent predictors of favorable treatment outcomes (FTOs).

3. Results

3.1. Clinical characteristics of elderly patients with TBI

In total, 493 patients were included in the analysis during the study period. Of these, 170 (34.5%) patients aged 65 years or older were categorized in the elderly group, and 323 (65.5%) patients

aged between 16 and 64 years were categorized in the younger group. The mean age of all patients was 54.3 ± 10.9 (16–96) years. Table 1 shows the clinical characteristics of the younger and elderly groups. The male to female ratio was greater in the younger group than in the elderly group. The mean GCS score at the ED was higher in the elderly group than in the younger group ($P = 0.038$). In addition, the mean time from head injury to arrival at the ED was significantly different between the two groups (younger: 3.5 ± 4.9 h vs. elderly: 5.7 ± 7.0 h; $P = 0.002$). The most common reason for injury in the elderly patients was a fall from a height of ≤ 1 m (43.5%), followed by motor vehicle accidents (17.6%), pedestrian accidents (8.2%), and a fall from a height of > 1 m (7.1%) (Fig. 1). The most common type of TBI was acute SDH in the elderly and epidural hemorrhage in the younger group (Table 2).

3.2. Treatment outcome

Thirty-nine (22.9%) patients in the elderly group were treated surgically. Surgical treatment included decompressive craniectomy in 11, craniotomy in 9, and drainage of hematoma through catheter insertion in 20 patients. One patient underwent a craniotomy followed by a decompressive craniectomy. There was no statistically significant difference in the 30-day mortality rate between the younger and elderly groups (14.9% vs. 19.4%; $P = 0.195$). Additionally, the rates of surgical treatment refusal were 1.6% and 7.7%, respectively, but the difference was not statistically significant ($P = 0.15$) (Table 3). Younger patients tended to accept surgical treatment, whereas elderly patients' caregivers tended to refuse surgical treatment because of their older age. There were no significant differences in treatment outcomes at 6 months between the younger and elderly patients ($P = 0.572$) when outcomes were dichotomized into favorable and unfavorable groups according to GOS-E score at 6 months after injury. Among the elderly patients,

Table 1
Patient characteristics.

Characteristics	Younger	Elderly	P Value
No. of patients (%)	323 (65.5)	170 (34.5)	
Sex			< 0.001
Male	261 (80.8)	97 (57.1)	
Female	62 (29.2)	73 (42.9)	
LOC	247 (76.5)	111 (65.3)	0.014
Antiplatelet/Anticoagulant agent	18 (5.6)	52 (30.6)	< 0.001
GCS (ED) (mean $\pm$ SD)	11.9 ± 3.9	12.7 ± 3.4	0.038
GCS (ED)			0.054
Severe (3–8)	73 (22.6)	26 (15.3)	
Moderate (9–12)	38 (11.8)	15 (8.8)	
Mild (13–15)	212 (65.6)	129 (75.9)	
AIS (Head)			0.343
Moderate-serious (2–3)	174 (54.2)	84 (49.7)	
Severe (≥ 4)	147 (45.8)	85 (50.3)	
ISS			0.604
Minor (1–8)	56 (17.3)	21 (12.4)	
Moderate (9–15)	101 (31.3)	53 (31.2)	
Serious (16–24)	74 (22.9)	56 (32.9)	
Severe (≥ 25)	92 (28.5)	40 (23.5)	
Treatment			0.883
Non-surgical treatment	247 (76.5)	131 (77.1)	
Surgical treatment	76 (23.5)	39 (22.9)	
Time from injury to ED (hours) (mean $\pm$ SD)	3.5 ± 4.9	5.7 ± 7.0	0.002

Categorical variable are presented as number (percentage).

LOC, loss of consciousness; GCS (ED), Glasgow coma scale (emergency department); SD, standard deviation; AIS, abbreviated injury scale; ISS, injury severity score; ED, emergency department.

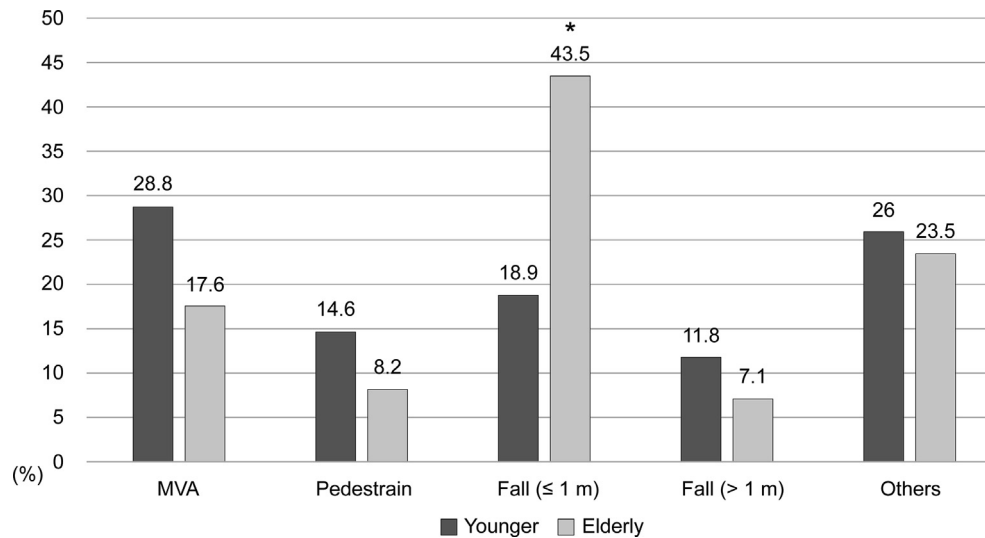


Fig. 1. Mechanism of injury. MVA, motor vehicle accidents. * $p < 0.05$; chi-square test.

Table 2

Types of intracranial hemorrhage.

Types	Younger (n = 323)	Elderly (n = 170)	P Value
Acute SDH	114 (36.2%)	134 (80.2%)	<0.001
EDH	196 (62.2%)	18 (10.8%)	<0.001
C-ICH	154 (48.9%)	67 (40.1%)	0.079
T-SAH	178 (56.5%)	82 (49.1%)	0.146
Others	18 (5.7%)	8 (4.8%)	0.682

Variables are presented as number (percent of cases).

SDH, subdural hemorrhage; EDH, epidural hemorrhage; C-ICH, contusion-intracerebral hemorrhage; T-SAH, traumatic-subarachnoid hemorrhage.

Table 3

Mortality and causes of death.

Variables	Younger (n = 323)	Elderly (n = 170)	P Value
Mortality	48 (14.9)	33 (19.4)	0.195
Cause of death			0.347
CNS-related	26 (53.2)	21 (63.6)	
Sepsis/DIC	17 (36.2)	12 (36.4)	
Cardiac origin	3 (6.3)	0	
Unknown	2 (4.3)	0	
Do not resuscitation			0.15
Refusal of surgical intervention	5 (1.6)	13 (7.7)	
Refusal of CPR	31 (9.5)	14 (8.5)	

Categorical variables are presented as number (percentage).

CNS, central nervous system; DIC, disseminated intravascular coagulation; CPR, cardiopulmonary resuscitation.

Table 4

Treatment outcome at 6 months according to age.

Outcome	Younger (n = 323)	Elderly (n = 170)	P Value	Age 65–74 (n = 87)	Age 75–84 (n = 67)	Age ≥ 85 (n = 16)	P Value
Unfavorable (GOS-E ≤ 4)	67 (20.7)	39 (22.9)	0.572	17 (19.5)	18 (26.9)	4 (25)	0.363
Favorable (GOS-E > 4)	256 (79.3)	131 (77.1)		70 (80.5)	49 (73.1)	12 (75)	

Categorical variables are presented as number (percentage).

GOS-E, Glasgow outcome scale–extended.

treatment outcomes were not significantly different according to age ($P = 0.363$) (Table 4).

3.3. Factors related to treatment outcome in elderly patients with TBI

Table 5 shows the results of stratification according to treatment outcome by GOS-E at 6 months in the elderly patients with TBI. There were significant differences in GCS score at the ED, ISS, pupillary reflex, and pure subdural hemorrhage (pSDH) between patients with FTO and those with unfavorable treatment outcome. However, there was no difference in the mean duration from head injury to arrival at the ED between the unfavorable and favorable groups. To evaluate treatment outcome according to the type of intracranial hemorrhage, patients who had pSDH without intraparenchymal injury were compared with other patients (“others” group) (Table 6). There were statistically significant differences in GCS score at the ED, mean GCS score, and abbreviated injury scale between the pSDH and “others” groups. Twenty-one of 62 elderly patients with pSDH underwent surgical treatment, which was a significantly different proportion compared with the “others” group. The surgical treatment included 6 patients with craniectomy, 2 patients with craniotomy, and 13 patients with drainage of hematoma through catheter insertion. Of these, 17 (81%) were patients with mild or moderate TBI with GCS scores of 9–15 points and a 0% mortality rate ($P = 0.003$); FTO was observed in 16 patients ($P = 0.012$). Patients with pSDH had a higher rate of FTO and a lower mortality rate than those in the “others” group.

We used multivariate logistic regression analysis to identify factors that were predictive of FTO in elderly patients with TBI. Mild or moderate GCS score at the ED (odds ratio [OR], 5.63; 95% confidence interval [CI] 1.156–27.433) and ISS <16 points (OR, 14.242; 95% CI 1.627–124.687) were significant factors associated with

Table 5
Comparisons of treatment outcome by GOS-E at 6 months in elderly patients with TBI.

Variables	Unfavorable (GOS-E ≤ 4) (n = 39)	Favorable (GOS-E > 4) (n = 131)	P Value
GCS (ED)			<0.001
Severe (3–8)	23 (59.0)	3 (2.3)	
Moderate (9–12)	8 (20.5)	10 (7.6)	
Mild (13–15)	8 (20.5)	118 (90.1)	
ISS			<0.001
Minor-moderate (1–15)	4 (10.3)	70 (53.4)	
Serious-severe (≥16)	35 (89.7)	61 (46.6)	
Normal pupillary reflex	1 (1.0)	19 (48.5)	<0.001
LOC	35 (90.9)	74 (56.7)	<0.001
pSDH	9 (23.1)	53 (40.5)	0.048
Surgical treatment	16 (41)	23 (17.6)	0.002
Time from injury to ED (hours)	5.4 ± 6.69	5.8 ± 7.16	0.739
GCS (D)	7.7 ± 3.95	13.9 ± 1.80	<0.001
GCS (D)	3.7 ± 3.28	14.7 ± 0.70	<0.001

Categorical variables are presented as number (percentage).

Continuous variables are presented as mean ± SD. TBI, traumatic brain injury; GOS-E, Glasgow outcome scale-extended; GCS (ED), Glasgow coma scale (emergency department); ISS, injury severity score; LOC, loss of consciousness; pSDH, pure subdural hemorrhage; ED, emergency department; SD, standard deviation; GCS (D), Glasgow coma scale (discharge).

Table 6
Characteristics and treatment outcomes for elderly patients according to intracranial hemorrhagic types.

Characteristics	pSDH (n = 62)	Others (n = 108)	P Value
Male	34 (54.8)	62 (57.4)	0.745
GCS (ED)			0.041
Severe (3–8)	6 (9.7)	20 (18.5)	
Moderate (9–12)	4 (6.4)	14 (13)	
Mild (13–15)	52 (83.9)	74 (68.5)	
GCS (ED) (mean ± SD)	13.52 ± 2.70	12.18 ± 3.73	0.001
AIIS (Head)			0.026
Moderate-serious (2–3)	24 (38.7)	61 (56.5)	
Severe (≥4)	38 (61.3)	47 (43.5)	
Surgical treatment	21 (33.9)	18 (16.7)	0.01
Treatment outcome			0.048
Unfavorable (GOS-E ≤ 4)	9 (14.5)	30 (37.8)	
Favorable (GOS-E > 4)	53 (85.5)	78 (72.2)	
Mortality	7 (11.3)	26 (24.1)	0.043

Categorical variables are presented as number (percentage).

pSDH, pure subdural hemorrhage; GCS (ED), Glasgow coma scale (emergency department); SD, standard deviation; AIIS, abbreviated injury scale; GOS-E, Glasgow outcome scale-extended.

Table 7
Factors for predicting favorable treatment outcome (GOS-E > 4) in elderly patients with traumatic brain injury.

Factors	Odds Ratio (95% Confidence Interval)	P Value
GCS (ED) >8	5.630 (1.156–27.433)	0.032
ISS <16	14.242 (1.627–124.687)	0.016
Normal pupillary reflex	51.543 (1.302–2040.324)	0.036
pSDH	4.159 (1.035–16.712)	0.045
Surgical treatment	0.876 (0.157–4.903)	0.88

Multivariate logistic regression analysis.

GOS-E, Glasgow outcome scale-extended; GCS (ED), Glasgow coma scale (emergency department); ISS, injury severity score; pSDH, pure subdural hemorrhage.

FTO. In addition, normal pupillary reflex (OR, 51.543; 95% CI 1.302–2040.324) and pSDH (OR, 4.159; 95% CI 1.035–16.712) were independent predictors associated with FTO (Table 7).

4. Discussion

The purpose of this study was to identify factors that were predictive of FTO to allow strategic treatment planning in elderly patients with TBI. Haring et al. [13] revealed that 950,132 TBI-related hospitalizations occurred among individuals ≥65 years of age in the United States, with 107,666 TBI-related deaths between 2000 and 2010. With the extension of life expectancy, elderly individuals are becoming more active in their fields of activity; moreover, interest in the health, care, and treatment of the elderly is growing. In the present study, the male/female ratio was greater in the younger group, while there was no statistically significant difference in the elderly group (Table 1). The number of female patients with TBI has risen rapidly among those >60 years of age, which may be due to an increasing life expectancy and social activities among women [14,15]. This was reflected in our study, where the proportion of female patients was higher in the elderly group than in the younger group. Song et al. [16] showed that there was increase in the incidence of TBI with increase in the age of patients, and that the proportion of middle-aged and older adults with TBI was higher than that given in reports from the United States.

Hawley et al. [17] reported that falls accounted for 85% of TBI cases. In Africa, falls accounted for 10% of TBI cases, but affected patients showed good recovery [18]. In the present study, the most common cause of TBI among the elderly was a fall from a height of ≤1 m, whereas motor vehicle accidents were the most common cause in the younger patients. In addition, the mean time from head injury to arrival at the ED was significantly different between the younger and elderly patients, but there were no statistically significant differences in the outcomes in elderly patients. This is probably due to the higher rate of mild or moderate TBI in elderly patients than in the younger group, as well as delayed symptom development caused by cerebral atrophy with large subdural space, reduced physical function, and deterioration of cognitive function (e.g., dementia) in elderly patients.

SDH is a common pathology in elderly patients. According to data collected by the United States Census Bureau, the incidence rate of SDH nearly doubles between 65 and 75 years of age and continues to increase in individuals >80 years of age [19]. As age increases, brain volume decreases and the capacity of the subdural space increases, leading to higher rates of SDH and intracerebral hemorrhage than epidural hematoma [20]. In the present study, 53.8% of the elderly patients who underwent surgical treatment had pSDH. In addition, 80.9% of the elderly patients who underwent surgical treatment for pSDH showed FTO. Elderly patients with pSDH had a significantly lower mortality rate and a higher rate of FTO at 6 months than those with other types of hemorrhage.

Mortality rates reported by studies on chronic SDH among the elderly range from 1.2 to 16.7%, [21,22] while those for acute SDH range from 35 to 88% [23–25]. This is an astounding change from the early 1980s and 1990s, when the mortality rates for people ≥65 years of age were 74–88% [24,26,27]. Kiboi et al. [28] reported that approximately 9% of patients >61 years of age in Africa had traumatic intracranial hematoma, and that age was a predictive factor for functional recovery. In addition, they demonstrated that early detection through brain computed tomography scan and surgical intervention were the main points of treatment in elderly patients. In the present study, the mortality rate was 19.4% in elderly patients, and a greater percentage of elderly patients compared to younger patients refused surgical treatment.

One hundred thirty-one (77.1%) of 170 elderly patients had FTO at 6 months after injury. Compared with several studies, [17,23,29,30] we suspect that the lower mortality rate and higher percentage of FTO in this study are due to the low proportion of patients with severe TBI (GCS 3–8). Initial GCS was a major prognostic factor, as reported by other previous studies in patients with severe head injury [31,32]. Utomo et al. [14] revealed that GCS score, brainstem injury, and systolic blood pressure were the most important factors in predicting outcome in elderly patients with moderate to severe (80%) TBI according to the Mayo criteria. In addition, Li et al. [10] reported that injury severities, pupillary responses, and computed tomography findings were other significant contributing factors in clinical decision-making. In the present study, multivariate analysis showed FTO in elderly patients with mild or moderate TBI, lower ISS, normal pupillary reflex, and pSDH. Prasad et al. [33] reported that the negative prognostic factors in elderly patients with TBI in developing countries were age >75 years, severe head injury, and acute SDH. These results on acute SDH may contradict our results because the previous study was conducted in a developing country. To improve the treatment outcomes of elderly patients with mild or moderate TBI, surgical treatment should be recommended for patients with pSDH without intraparenchymal injury.

There were several limitations to this study. First, patient selection may have been biased because of the retrospective design. A prospective randomized control trial is needed to better determine the therapeutic effects of surgical strategies in elderly patients. Second, the study was conducted with a small number of patients at a single institute. In addition, more patients had mild TBI than moderate or severe TBI. Third, the evaluation of treatment efficacy was only compared with functional outcome using the GOS-E at 6 months after injury. Finally, in order to accurately assess the effectiveness of surgical treatment of head injury, it is necessary to analyze a broader group of patients, except those with other major organ injuries.

5. Conclusion

We presented the characteristics and treatment outcomes of elderly patients with TBI in this single-center study. Compared to younger patients, elderly patients did not show a significant difference in 30-day mortality rate and GOS-E at 6 months after injury. Higher GCS score at the ED, lower ISS, normal pupillary reflex, and pSDH without intraparenchymal injury were significant predictive factors for FTO in elderly patients with TBI. The results of this study suggest that elderly patients with TBI with pSDH, if provided with appropriate treatment strategies, including surgery, can achieve FTO following TBI. Based on the restrictions of recovering from surgical treatment, a more minimally invasive approach is needed.

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Declaration of Competing Interest

None.

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